

🌀 M003 Extension — The Loop Pattern (Intermediate)

Topic: One loop, three counters (x, y, z) · **Course:** 3D Coordinates · **Level:** Extension (Intermediate, after Week 6) · **Time:** about 40 minutes

Until now you placed and filled blocks by choosing **every** coordinate **by hand**. This time **one loop** builds all **7** obsidian walls for you. The loop keeps three counters — **x, y, z** — and nudges each one by a fixed **step** every pass. That is an **algorithm**: a short list of steps a computer repeats. Load the world, type **SPIRAL** in the chat, and watch 7 walls grow into a pattern.

```
x = 14
y = 0
z = 2
for step in range(7):
    fill OBSIDIAN from (14, 0, z) to (x, y, z)
    x = x - 2
    y = y + 1
    z = z + 2
```

 **Big idea:** the **fill** line **never changes** — it is the same seven times. The walls come out **different** only because **x, y, z change** between passes.

Each pass the new wall is **2 wider** (x), **1 taller** (y), and **2 deeper** (z) than the last.

- Stand on your **home spot** (red block) every run.
- 🧠 Every wall is **obsidian**.

1 · Predict 🧠

Trace the loop. For each iteration, write the **START** values (what x, y, z hold when the pass begins — the numbers the **fill** line uses) and the **END** values (after the three update lines run). Iteration 0 is done for you. The **END of one row is the START of the next**.

Iteration	START x	START y	START z	END x	END y	END z
0	14	0	2	12	1	4
1						

2						
3						

(The pattern keeps going to iteration 6.)

The `fill` line runs *before* the updates. Which `x`, `y`, `z` does iteration 0's wall use? Circle one: (14, 0, 2) · (12, 1, 4)

Which counter makes each wall taller? Circle one: `x` · `y` · `z`

`range(7)` builds how many walls — and why does each wall come out bigger even though the `fill` line is never edited? (1–2 sentences)

2 · Spot the Bug 🐛

Each loop is broken. Read what it should do, fix it, and say why.

Bug A — a counter that never steps. The `z = z + 2` line was deleted, so `z` stays **2** the whole time.

```
for step in range(7):  
    fill OBSIDIAN from (14, 0, z) to (x, y, z)  
    x = x - 2  
    y = y + 1
```

What do you see instead of a pattern? Circle one: all 7 walls stacked at `z = 2` · 7 walls spread out

Write the missing line and where it goes:

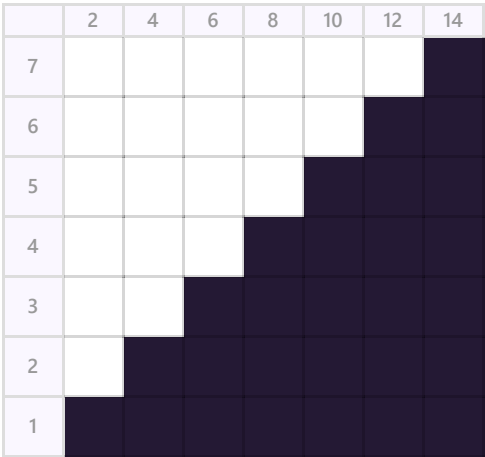
Bug B — wrong step size. A friend changed one line to `z = z + 1`. It runs, but the 7 steps merged into one solid slope with no gaps between them.

The walls are each 1 block thick. Explain why `z = z + 2` leaves a gap but `z = z + 1` makes the steps touch, then write the fix.

3 · Show Your Work 📷 🎥

Load the world. Stand on your home spot. Type `SPIRAL` in the chat and watch the loop build all 7 walls. Walk around the pattern — see the tiny 1-block step out front and the tall 7-high wall at the back.

 The finished pattern — side view (y ↑ up, z → deeper)



Reading left to right, the steps climb 1, 2, 3, 4, 5, 6, 7 blocks high — one taller each pass, because `y = y + 1` runs every loop.

Modify challenge. Change **one** thing and **predict the new pattern before you run it** — for example change `range(7)` to `range(10)` , or make the step `y = y + 2` (steeper). Write what will move, then build it and check.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what the loop does and what each counter changes.

2 · Your build. Point the camera at the pattern. Name the smallest and biggest step.

Fill the blanks:

Today I built ____.

I built it using this code: ____.

In this code I used ____.

The hardest part was ____.

That part was hard because ____.

The most fun part was _____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit 

Send this worksheet + your **one** video to teacher on KakaoTalk.